

New Data Model Results

Training dataset: dec2025rb2r2_updated.txt

Testing dataset: feb2026rb2r2_updated.txt

Voltage Results:

Model	MAE	RMSE	R ²
DeepNN	0.0684	0.1117	0.8899
DNN	0.0890	0.1690	0.7480
LSTM	0.0684	0.1372	0.8339

Power Results:

Model	MAE	RMSE	R ²
DeepNN	1.5718	4.0836	0.99916
DNN	1.2301	3.0590	0.99953
LSTM	6.7059	38.2457	0.92633

1. Voltage Prediction

DeepNN performs the best with the highest R² (approx. 0.89), indicating it effectively captures nonlinear relationships in the data. LSTM shows comparable performance but does not provide significant improvement suggesting that temporal dependencies in voltage are limited. DNN performs the worst, indicating that a shallow model is insufficient.

Overall, voltage prediction benefits from nonlinear modeling but does not strongly depend on temporal information.

2. Power Prediction

DNN achieves the best performance with an R² of approx. 0.9995, followed closely by DeepNN. LSTM performs significantly worse, indicating that sequence modeling is not suitable for this task.

This is because power is governed by instantaneous relationships rather than temporal dependencies. Therefore, simpler feedforward models outperform more complex sequence-based models.